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Research Article

Photo-Initiated Nickel Catalysis (PiNiC): Unmasking Dimethylnickel with Light

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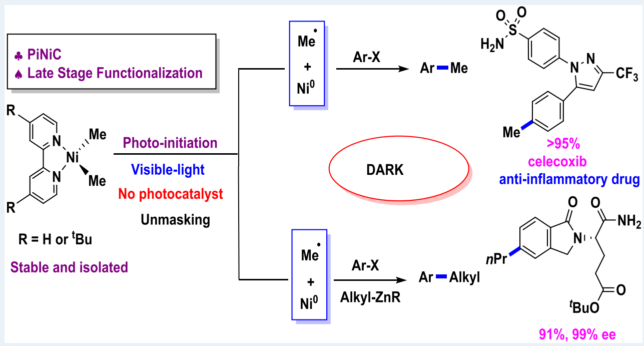

ABSTRACT: Nickel catalysis has garnered interest in mainstream organic synthesis for constructing chemical bonds and has been merged with photoredox catalysis and electrocatalysis. Despite its success and increasing adoption by academia and industry, the mechanism by which Ni catalyzes C−C cross-coupling reactions and how the precatalyst is activated remains unclear. An exploration of stabilized dimethylNi^{II} complexes, [(dtbbpy)Ni(Me)$_2$] (dtbbpy = 4,4′-di-*tert*-butyl-2,2′-bipyridine, **1**) and [(bpy)Ni(Me)$_2$] (bpy = 2,2′-bipyridine, **2**), as Ni-precatalysts for promoting $C_{sp}{}^2$−$C_{sp}{}^3$ cross-coupling revealed that they readily undergo photo-induced Ni−C bond homolysis upon exposure to visible light and are a source of Ni^0 species. Photo-initiation of dimethylNi^{II} complexes **1** and **2** produced active Ni^0 and reactive methyl radicals, which were harnessed to promote methylative and other $C_{sp}{}^2$−$C_{sp}{}^3$ cross-coupling reactions. The mechanistic studies conducted to illuminate aspects of the process include a competition reaction, Ni^{II}ArBr-complex synthesis, light-on and light-off studies, density functional theory (DFT) calculations of the bond dissociation free energies (BDFE), as well as control experiments with a Ni^0-precursor. We further demonstrated the robustness and utility of dimethylNi^{II} as a Ni^0-precursor in the late-stage installation of methyl, CD$_3$, alkyl, and benzyl groups on aryl bromides that have potential value in drug design and optimization. This work sheds light on the activation mechanism of Ni-catalyzed cross-coupling of aryl halides with organozinc reagents when a Ni^{II}-precatalyst is employed and the critical role that visible light plays in the process.

KEYWORDS: *Ni catalysis, photo-initiation, cross-coupling, dimethylnickel, methylation, late-stage-functionalization*

INTRODUCTION

Over the past 2 decades, the merger of nickel and photoredox catalysis, referred to as metallaphotoredox catalysis, has become an attractive cross-coupling platform for chemical synthesis, and this dual catalytic platform has unveiled multiple chemical transformations.[1−5]

The development of a robust $C_{sp}{}^2$−$C_{sp}{}^3$ cross-coupling process amenable to late-stage functionalization attracted our attention since it could help to meet the growing demand for molecular editing to access molecules with a higher fraction of sp^3 (Fsp^3) centers.[6−10] The so-called "magic methyl" effect has been recognized by medicinal chemists as one approach to enhancing the pharmacological properties of a molecule that includes modulating the potency and/or metabolic stability of drug molecules; hence, our interest in developing a robust methylative cross-coupling methodology.[11−14] Only two metallaphotoredox methylation processes have been reported in the literature, both of which are Ni-catalyzed cross-coupling methods that employ an iridium-based photoredox catalyst.[14a,15]

Metal-catalyzed cross-coupling reactions are effective protocols for the construction of C−C bonds.[16−19] Conventional metal-catalyzed cross-coupling methods would be considered first by synthetic chemists for the late-stage coupling of **I** and **II** to access a molecule of type **III**, which introduces several sp^3 centers (Figure 1A). However, the need for a high temperature and expensive, exotic ligands can be practical limitations that are exacerbated by difficult precatalyst activation, catalyst arrest, or the slow reactivity of the two-electron, transmetalating $C_{sp}{}^3$ organometallic reagents.[16−19] As a result, there has been a rapid shift to taking advantage of single-electron processes and techniques such as metallaphotoredox catalysis and electrochemistry/electrocatalysis, which exploit the unique reactivity of Ni-complexes.[1−5,20−28] Single-electron

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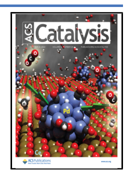

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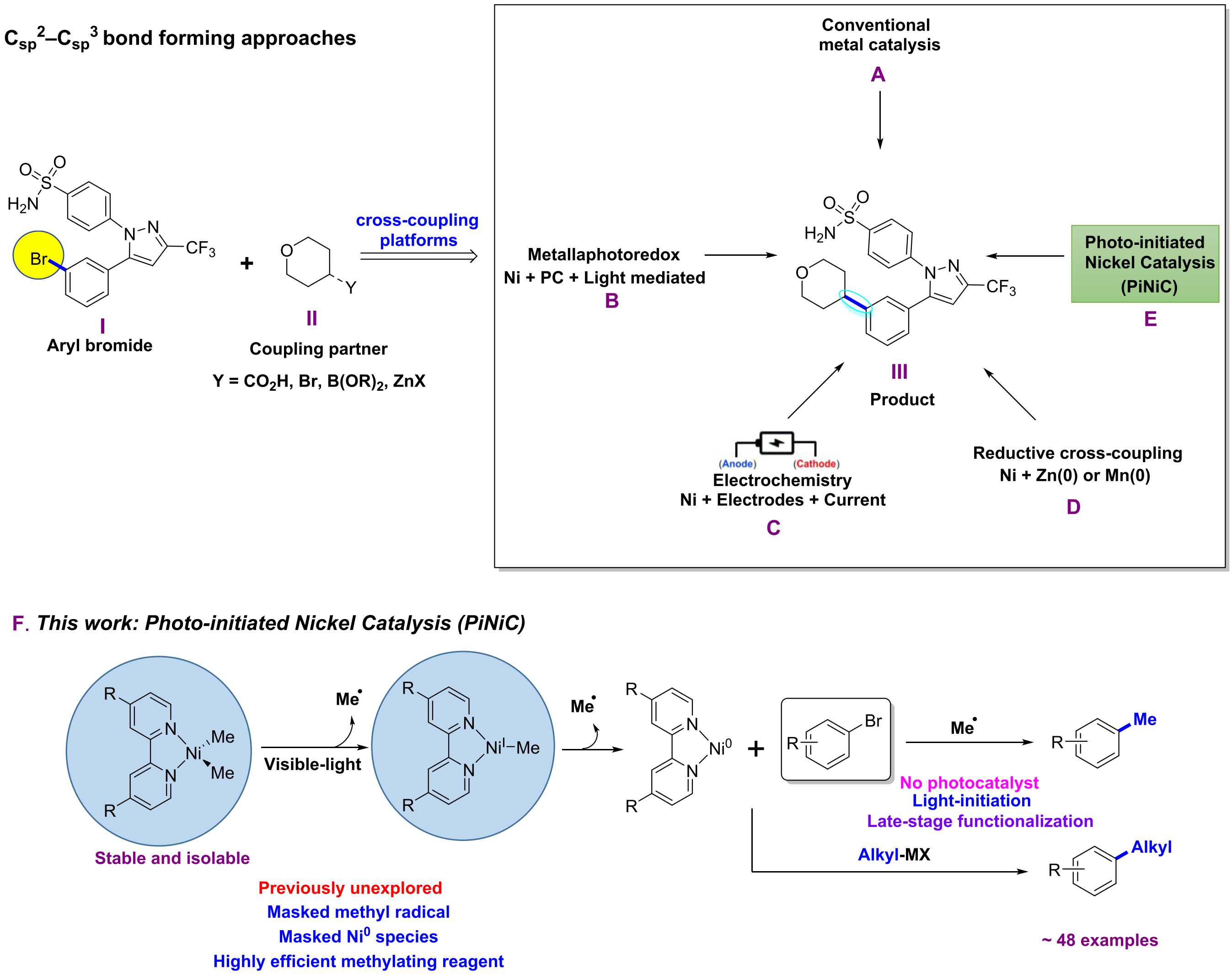

Figure 1. Cross-coupling approaches and reaction design: Four established approaches for the cross-coupling of aryl bromide **I** and coupling partner **II** to access product **III** using Ni catalysis: Conventional methods (A); metallaphotoredox catalysis (B); electrocatalysis (C); Zn(0) or Mn(0)-mediated reductive cross-coupling strategies (D). An emerging cross-coupling platform, photo-initiated nickel catalysis, PiNiC (E). This work: A photo-initiated, nickel-catalyzed, $C_{sp}{}^2$–$C_{sp}{}^3$ cross-coupling reaction promoted by stabilized dimethylNi^{II} via photo-induced Ni–C homolysis (F).

cross-coupling processes that encompass metallaphotoredox catalysis (Figure 1B) and electrochemistry (Figure 1C) are attractive, in part, due to the mildness of the reaction conditions and have led to the development of highly chemoselective methodologies amenable to late-stage functionalization.[20−34] Metallaphotoredox catalysis has combined the efficiency of Ni catalysis for bond formation with the electron- and energy-transfer capacity of photoredox catalysis.[1,34] However, the high cost of photoredox catalysts and their limited availability for scale-up campaigns have slowed down its adoption in process development. Although electrochemically driven, Ni-catalyzed transformations have been performed on a large scale,[25,30] electrochemical processes are seeing a slower adoption by industry despite their considerable potential. Ni-catalyzed cross-electrophile coupling mediated by either Zn(0) or Mn(0) under reducing conditions is an alternate cross-coupling platform but requires the use of stoichiometric amounts of a reductant (Figure 1D).[35,36] To circumvent the limitations of established methods, we considered a photo-initiated Ni catalysis (PiNiC) platform (Figure 1E) for cross-coupling as a cost-effective and alternative approach to metallaphotoredox and electrocatalytic processes. We reasoned that the light-mediated initiation of a metal precatalyst could preclude the use of a photoredox catalyst for both single- and two-electron cross-coupling techniques. Against that backdrop, we sought a stable Ni-complex that could promote C−C cross-couplings as a prelude to establishing a PiNiC platform.

In 1966, Wilke and Herrmann reported the first bipyridine-stabilized dialkylnickel complexes, including $[(bpy)Ni(Me)_2]$ (**2**).[37] The single crystal X-ray structure of **2** was solved by Holm in 1995, showing that the planar complex is stable at room temperature.[38] Detailed studies of the UV−vis spectra of the dimethylNi^{II} complexes **1** and **2** were later reported independently by Vicic and Klein.[39,40] The experimental UV−vis spectra of **1** and **2** are of significance to our investigation since their metal−ligand charge transfer (MLCT) bands appear in the visible-light region.[39,40] Surprisingly, neither complex has been explored for its potential in photochemical reactions and, as such, their viability in visible-light photo-

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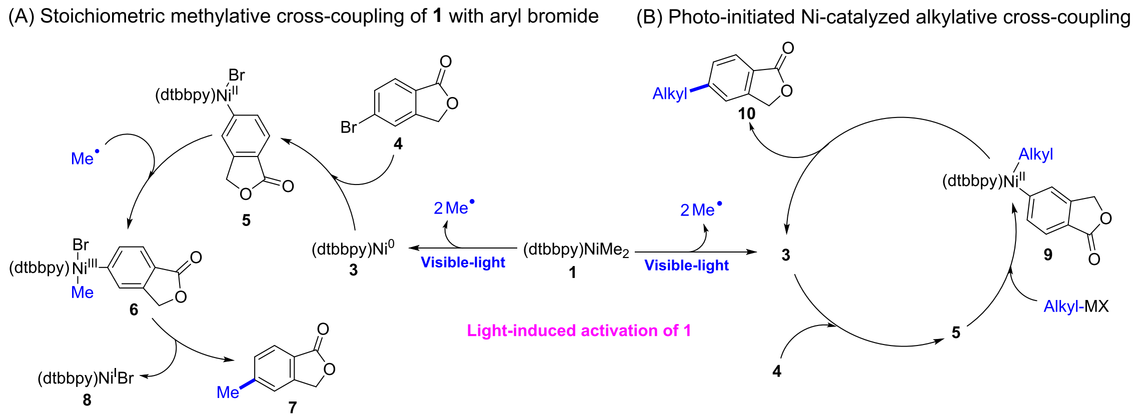
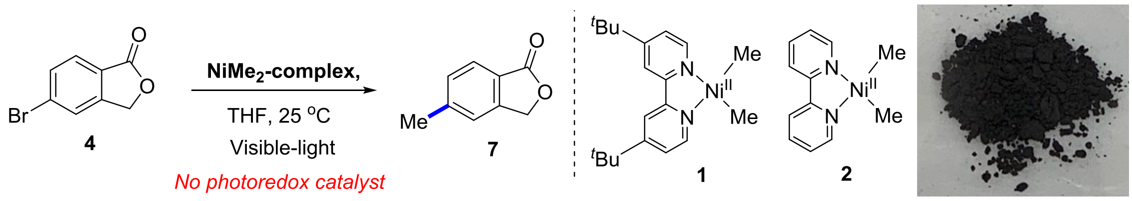


Figure 2. Reaction design and proposed mechanistic underpinning: Proposed mechanism for the stoichiometric, methylative cross-coupling of **1** with an aryl bromide **4** (A). Photo-initiated, Ni-catalyzed $C_{sp}{}^{2}-C_{sp}{}^{3}$ cross-coupling (B).

Table 1. Results of a Study of the Stoichiometric and Catalytic Competence of 1 and 2 to Mediate Cross-Coupling Reactions[a]



entry	$NiMe_2$	conditions	% conversion[b]
1	**1** (1 equiv)	without light	N.D.[g]
2	**1** (1 equiv)	blue light (450 nm)	99 (95)[b]
3	**2** (1 equiv)	blue light (450 nm)	99[b]
4	**1** (1 equiv)	violet light (400 nm)	99
5	**1** (5 mol %)	2 equiv Me_2Zn[c]	99
6	**1** (5 mol %)	2 equiv MeZnCl[c]	95
7	**2** (5 mol %)	2 equiv Me_2Zn[c]	99
8	**1** (5 mol %)	2 equiv Me_3Al[c]	>95%
9	**1** (5 mol %)	1 equiv MeMgCl[c]	N.D.
10	**1** (5 mol %)	with oxygen, no light[d]	N.D.
11	**1** (5 mol %)	with oxygen, irradiated with blue light	99
12	**1** (5 mol %)	2 equiv Me_2Zn[e]	95
13	**2** (5 mol %)	2 equiv Me_2Zn[e]	95
14	**1** (5 mol %)	DMF[e,f]	99
15	**1** (5 mol %)	DMA[e,f]	99
16	**1** (5 mol %)	DMSO[e,f]	90
17	**1** (5 mol %)	MeCN[e,f]	35

[a]The picture of bench-stable **2** is shown above. Reaction protocols: stoichiometric conditions: a mixture of **1** or **2** (0.5 mmol), **4** (0.5 mmol) in THF (0.1 M) was irradiated with 34−50 W blue LEDs at 25 °C under an atmosphere of N_2 for 16−18 h. Catalytic conditions: a mixture of **1** or **2** (0.025 mmol, 5 mol %), **4** (0.5 mmol), Me_2Zn (1.0 mmol, 2 equiv, 1 M solution), and THF (0.1 M) was irradiated with 34−50 W blue LEDs at 25 °C under an atmosphere of N_2 for 16−18 h. [b]UPLC conversion; isolated yield in parentheses. [c]This reaction was irradiated with 34−50 W blue LEDs at 25 °C for 16−18 h. [d]Wrapped with aluminum foil to avoid light penetration. [e]This reaction was irradiated with 34−50 W blue LEDs at 25 °C for only 5 min, after which the light was turned off and then stirred for 16−18 h. [f]Solvent used instead of THF. [g]N.D. means no desired product was observed.

catalysis has been enigmatic. To this end, we questioned whether a ligand-stabilized dimethylNi^{II} catalyst could mediate $C_{sp}{}^{2}-C_{sp}{}^{3}$ cross-coupling reactions under irradiation with visible light. Specifically, it was hypothesized that visible-light absorption by **1** and **2** had the potential to induce Ni−C bond homolysis to generate a methyl radical and a Ni^{I}Me-species.[41,42] Although not apparent, a second Ni−C bond homolysis reaction of the resulting Ni^{I}Me-species could also occur to generate an active Ni^{0}-species (Figure 1F). Mechanistic studies by Doyle and co-workers[43,44] showed that photo-excited (dtbbpy)Ni^{II} aryl halide complexes could undergo Ni−aryl homolysis to generate aryl radicals and Ni(I), thus providing credence to our hypothesis. Given that alkyl radicals are also known to rapidly combine with $ArNi^{II}Br$, we hoped to harness the reactivity of both the methyl radicals and the Ni^{0} produced from dimethylNi^{II} **1** and **2**. We recognized that the relative rates of the photo-induced homolysis of the Ni−C bond in $Ni^{II}Me_2/Ni^{I}Me$, the homodimerization of methyl radicals to form ethane, the oxidative addition of Ni^{0} to an aryl bromide, and methyl radical capture by $ArNi^{II}Br$ would be crucial to the execution of this new cross-coupling strategy.[1] If successful, **1** and **2** could serve as attractive sources of methyl

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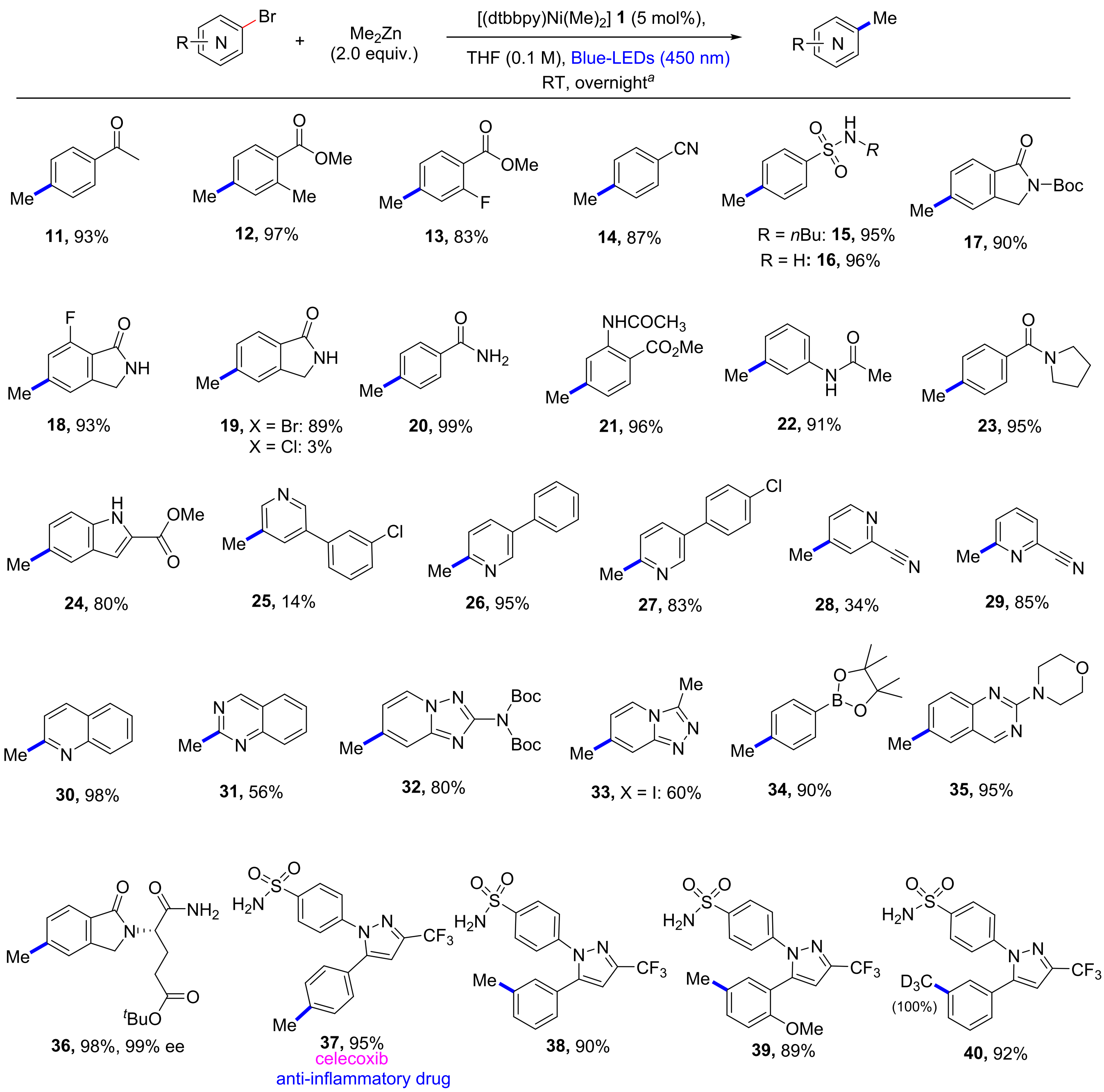
Scheme 1. Scope Studies of the Methylative Cross-Coupling of Aryl Bromides Catalyzed by 1



[a]A mixture of **1** (0.025 mmol, 5 mol %), aryl bromide (0.5 mmol), Me_2Zn (1.0 mmol, 2 equiv, 1 M solution in hexane), and THF (0.1 M) was irradiated with 34−50 W blue LEDs at 25 °C for only 5 min, after which the light was turned off and then stirred for 16−18 h.

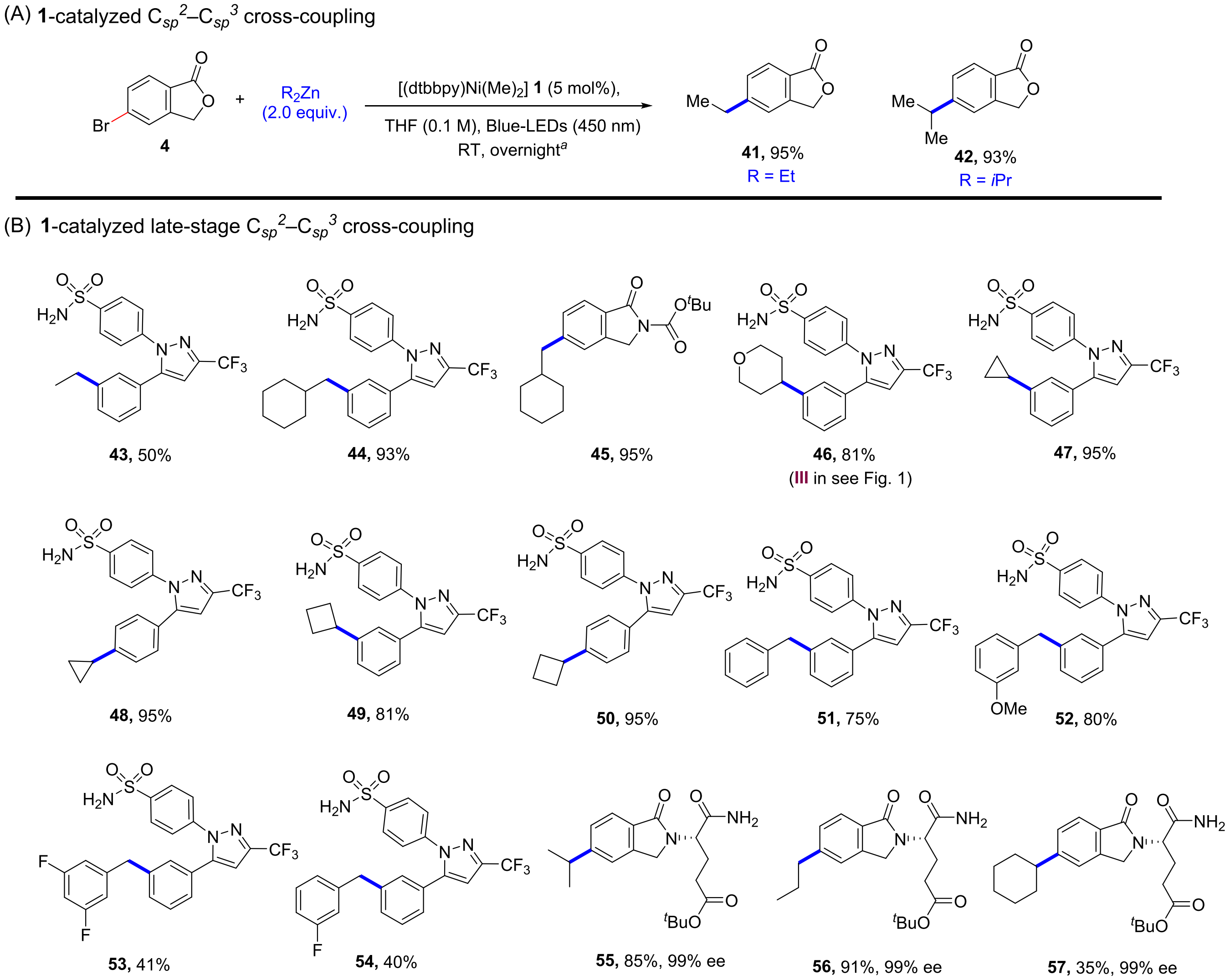
radicals, thereby providing an alternative option to the commonly used methylating agents such as the explosive *tert*-butylperoxide.[13]

Herein, we explored the feasibility of activating dimethylNi^{II} complexes by visible-light-induced Ni−C bond homolysis, which led to the development of a generically useful $C_{sp}{}^2$−$C_{sp}{}^3$ cross-coupling methodology and photo-initiated, Ni-catalyzed cross-coupling reactions.

A detailed description of the proposed mechanism for promoting $C_{sp}{}^2$−$C_{sp}{}^3$ cross-coupling with stoichiometric and catalytic amounts of **1** is outlined in Figure 2. The process commences with the photo-initiation of **1** under irradiation with blue-LED light (Figure 2A). Upon irradiation, **1** would undergo a MLCT excitation,[39,40] followed by two consecutive Ni−C bond homolyses to produce two methyl radicals and a (dtbbpy)Ni^0 species **3**. Thus, **1** is perceived as a masked source of methyl radicals and a Ni^0 species **3**. The Ni^0 species **3** generated in this reaction can undergo oxidative addition into an aryl bromide **4** to afford the Ni^{II} adduct **5**.[1,11,16] Interception of the methyl radical by **5**[1−5] would then generate an organometallic Ni^{III} adduct **6**, which upon reductive elimination, would deliver the methylated product **7**, with concomitant release of Ni^I bromide **8**.[1−4]

We also reasoned that upon the photo-initiation, **1** could be used catalytically to promote other $C_{sp}{}^2$−$C_{sp}{}^3$ cross-couplings (Figure 2B). If an alkyl transmetalating reagent is present,

Scheme 2. $C_{sp}{}^2$−$C_{sp}{}^3$ Cross-Coupling Catalyzed by 1 in a Late-Stage Functionalization Protocol



[a]A mixture of **1** (0.025 mmol, 5 mol %), aryl bromide (0.5 mmol), R_2Zn or RZnX (1.0 mmol, 2 equiv, 0.5−1 M solution), and THF (0.1 M) was irradiated with 34−50 W blue LEDs at 25 °C for only 5 min, after which the light was turned off and then stirred for 16−18 h. No product was obtained in the absence of light irradiation.

transmetalation with **5** would produce a Ni^{II} intermediate **9**, thus competing with the methyl radical trapping pathway. The reductive elimination of $ArNi^{II}$alkyl intermediate **9**[45] would forge the cross-coupled product **10**, with the concomitant regeneration of the (dtbbpy)Ni^0 species **3**.[45] It is noteworthy that the oxidation potential of **1** (E_{ox} = −0.60 V vs Fc^+/Fc in THF)[39,40] precludes a single-electron transfer (SET) to an aryl bromide (E_{red} of PhBr = −2.44 V vs SCE in DMF).[46]

■ RESULTS AND DISCUSSION

We began by preparing the stabilized dimethylNi^{II} complexes **1** and **2**, which were isolated as greenish-colored dark solids (see the Supporting Information).[37−40] Having isolated **1** and **2**, their stoichiometric and catalytic competence in cross-coupling reactions in the absence of a photocatalyst was studied (Table 1). When a stoichiometric amount of $[(dtbbpy)Ni(Me)_2]$ **1** was reacted with 5-bromophthalide **4** in the absence of light, no detectable reaction occurred (entry 1). As anticipated by our mechanistic reasoning, exposure of the reaction mixture to blue light (450 nm) gave the methylated cross-coupled product **7** in a 95% isolated yield (entry 2). Interestingly, $[(bpy)Ni(Me)_2]$ **2** is also competent at mediating the methylative cross-coupling of **4** to **7** under blue-light irradiation (entry 3). Irradiation with the high-energy violet light (400 nm) equally promoted the methylative cross-coupling reaction (entry 4). To the best of our knowledge, these results represent the first exploration of dimethylNi^{II} complexes in light-induced cross-coupling reactions.

Next, we focused on developing a catalytic version of this cross-coupling protocol. To that end, we screened several available methyl transmetalating reagents using 5 mol % of **1** or **2** to promote the methylative coupling reaction. While solutions of Me_2Zn, MeZnCl, and Me_3Al were immediately identified as suitable methyl-containing organometallic reagents for promoting the methylation reaction, MeMgCl was found to be ineffective (entries 5−9). We have previously reported that Ni-metallaphotoredox catalysis is tolerant of the presence of molecular oxygen;[47] hence, the reactivity of the dimethylNi^{II} complexes in the presence of molecular oxygen (air) was examined. While the exposure of the reaction mixture

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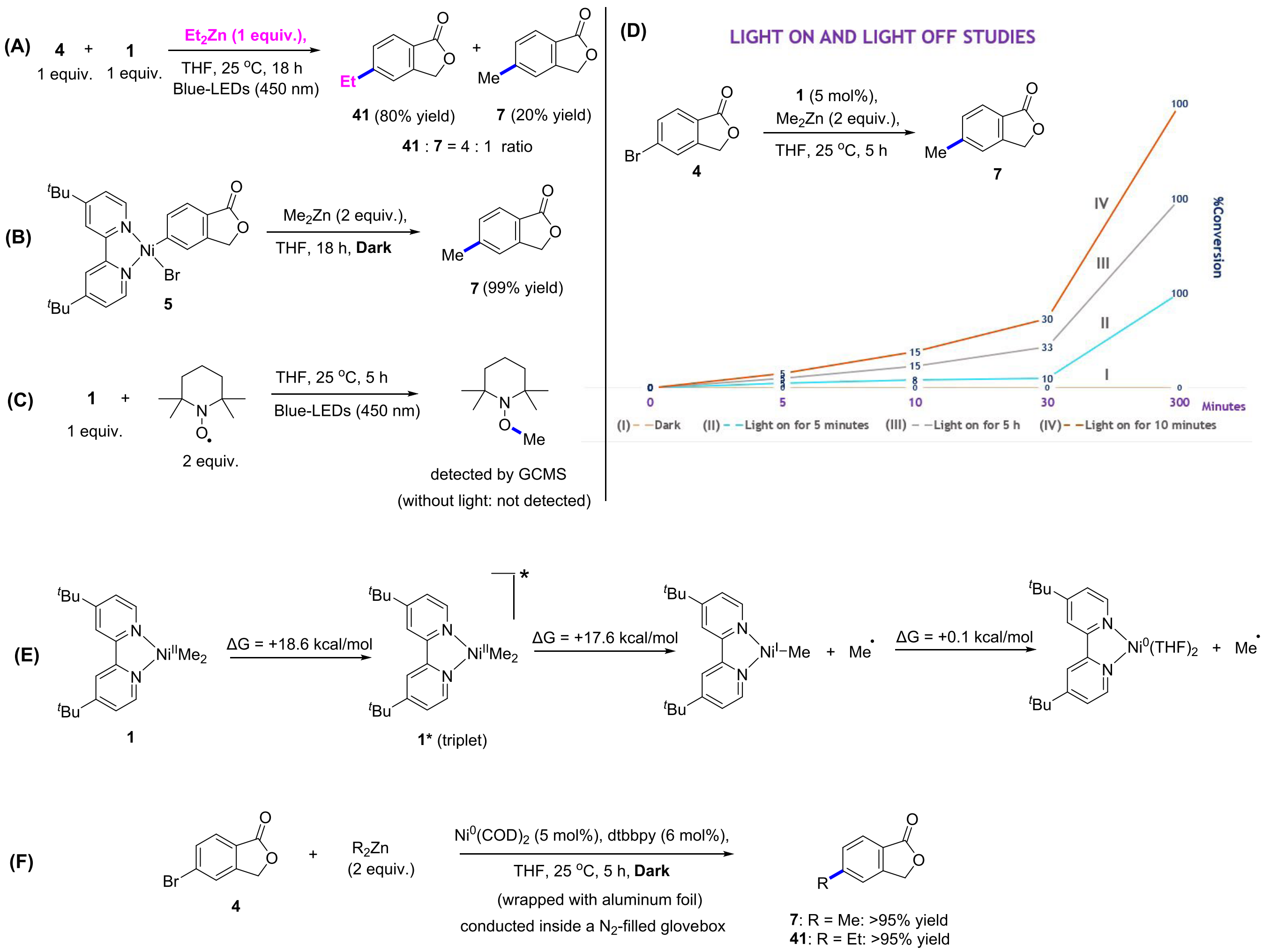

Figure 3. Mechanistic studies: (A) Probing the involvement of $Ni^{I}Me$ by cross-coupling **4** and Et_2Zn with a stoichiometric amount of **1**. (B) Reaction of the oxidative addition adduct **5** with Me_2Zn in the dark. (C) Methyl radical trapping experiment with TEMPO. (D) Light-on and light-off studies showing that activation of **1** is initiated by light in the Ni-catalyzed, $C_{sp}{}^{2}-C_{sp}{}^{3}$ cross-coupling protocol. %Conversion was determined by LCMS. In the light-off protocol, the vial was wrapped with aluminum foil and shielded completely from light. Light-on refers to a process in which the reaction was irradiated with 34−50 W blue LEDs (450 nm). Light-on and light-off experiments were conducted at 25 °C. (E) Estimation of (T_1-S_0) free energy gap and BDFE (bond dissociation free energies in kcal/mol) of Ni−C bond of $[(dtbbpy)Ni(Me)_2]$ (**1**) using DFT calculations. (F) Control reactions with a Ni(0)-source in the dark and carried out inside a nitrogen-filled glovebox.

to air in the dark resulted in the absence of cross-coupled product (entry 10), the reaction proceeded with quantitative conversion to **7** when exposed to both air and light (entry 11), suggesting that the reaction is tolerant to molecular oxygen (air) and that light is crucial for the activation of the dimethylNi^{II} catalyst. The reaction catalyzed by either **1** or **2** proceeded with quantitative conversion to **7** when irradiated with blue light for only 5 min and then stirred without light overnight (entries 12 and 13).

Solvent screening revealed that the reaction can be performed in a range of polar solvents, including DMF, DMA, and DMSO, without loss of efficiency (entries 14−16). However, a low conversion to **7** was observed when the reaction was performed in MeCN (entry 17).

Although dimethylNi^{II} complexes **1** and **2** mediated the methylation of **4** with similar efficiency, we chose to focus further study on the scope of the process using **1**, which bears dtbbpy, a widely employed ligand in Ni catalysis. With a new methylation methodology established, we sought to evaluate the generality of this protocol for the methylative cross-coupling of a variety of aryl bromides, including the late-stage functionalization of drug-like molecules. A broad range of electronically and structurally differentiated aryl bromides was found to couple efficiently using this protocol (Scheme 1). A range of functionalities on the aryl bromide are well tolerated that encompasses electron-withdrawing and electron-donating substituents, including a ketone, an ester, methyl, fluorine, cyano (**11**−**14**, 83−97% yield), sulfonamides (**15** and **16**, 95 and 96% yield, respectively), protic and aprotic amides, and both acyclic and cyclic amides (**17**−**23**, 89−99% yield). While 5-bromoisoindolin-1-one readily participated in the cross-coupling reaction to afford **19**, 5-chloroisoindolin-1-one gave only traces of the cross-coupled product.[17] Perhaps most notably, heteroaryl bromides, including unprotected bromoindole (**24**, 80% yield), bromopyridines (**25**−**29**, 14−95% yield), bromoquinoline (**30**, 98% yield), bromoquinazoline (**31**, 56% yield), and bromobenzotriazoles (**32** and **33**, 80 and 60% yield, respectively), which are traditionally challenging substrates in metal-catalyzed cross-couplings,[18] delivered the methylated heteroarenes in good to excellent yields. Although bromoar-

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enes appear to be more suitable as the electrophilic coupling partner, the iodo-subsituted heteroarene 7-iodo-3-methyl-[1,2,4]triazolo[4,3-*a*]pyridine underwent cross-coupling using this protocol to afford **33**. The ability to engage an aryl bromide bearing a boronate (Bpin) (**34**, 90% yield) underscores the mild and selective nature of the protocol.[48] Other bromoarenes with sensitive functional groups were also selectively methylated in excellent yields (**35** and **36**, 95 and 98% yield).

The robustness of the dimethylNi^{II} **1**-mediated protocol was further demonstrated with the late-stage installation of a methyl substituent to pharmaceutical precursors bearing aryl bromides. The marketed anti-inflammatory drug celecoxib (**37**) and its analogues **38** and **39** were readily accessed in high yields from their bromide-substituted precursors in 89−95% yields. Given the recent interest in deuterium labeling of drug molecules,[14b] we examined the potential to install a CD_3 group in a pharmaceutical drug precursor using the late-stage protocol. Using $(CD_3)_2Zn$, the deuterated **40** was obtained in a high 92% yield from the bromo-substituted precursor.

We also examined if **1** might be exploited in other Ni-catalyzed $C_{sp}{}^2-C_{sp}{}^3$ cross-coupling processes. In line with the mechanism proposed in Figure 2B, we investigated the cross-coupling of **4** with Et_2Zn using catalytic amounts of **1** (Scheme 2A). Interestingly, the ethylated product **41** was obtained from the reaction in a process that proceeded with full conversion without a trace of the methylated product **7**. Similarly, the cross-coupling of **4** and $i$$Pr_2Zn$ catalyzed by **1** gave only the isopropylated product **42** in 95% yield, results that collectively demonstrate, to the best of our knowledge, the first dimethylNi^{II}-mediated aryl−alkyl cross-coupling reactions. Inspired by these results, the scope of the alkyl moiety in this reaction was further explored with a particular focus on its potential for late-stage functionalization by $C_{sp}{}^2-C_{sp}{}^3$ cross-coupling. To this end, the late-stage installation of ethyl (**43**, 50% yield), cyclohexylmethyl (**44** and **45**, 93 and 95% yield, respectively), tetrahydropyran (**46**, **III** in Figure 1, 81% yield), cyclopropyl (**47** and **48**, 95% yield), cyclobutyl (**49** and **50**, 81 and 95% yield, respectively), and electronically diverse benzyl groups (**51**−**54**, 40−80% yield) were effectively catalyzed by **1** (Scheme 2B). We also applied this protocol to mediate the cross-coupling reaction of advanced enantiopure aryl bromides that have chiral centers that are susceptible to racemization. Isopropyl, propyl, and cyclohexyl groups were all successfully installed to afford **55**−**57** (35−91% yield) with excellent selectivity and with retention of the chirality (>99% ee).

Finally, we performed detailed experiments designed to probe the mechanisms proposed in Figure 2 and to examine the possible involvement of Ni^{I}Me species in the process (Figure 3). To this end, a 1:1:1 mixture of **1**, **4**, and Et_2Zn in THF was irradiated with blue light (Figure 3A). If the cross-coupling reaction was mediated entirely by a Ni^{I}Me species, only the methylated product **7** should be produced. However, a 4:1 ratio of the ethylated (**41**) and methylated (**7**) products was observed (Figure 3A). This result suggests that while the Ni^{I}Me species may contribute to the cross-coupling event via a Ni^{I}/Ni^{III} reductive elimination pathway to give **7** (Figure 2A), the Ni^{II}/Ni^{0} pathway that would give **41** appears to be dominant (Figure 2B). The process that proceeds via transmetalation appears to be faster than the one mediated by trapping of the methyl radical (see Figure 2A vs Figure 2B). To probe the intermediacy of **5** in the cross-coupling reaction (see Figure 2), we synthesized this molecule and used it for a control experiment. The proposed mechanistic pathway illustrated in Figure 2B suggests that light is only required for the initiation process and that the catalytic cycle operates without the need for light. Therefore, with **5** in hand, we performed a stoichiometric reaction with Me_2Zn in the dark that, consistent with the proposed mechanism, provided the cross-coupled product **7** in quantitative yield (Figure 3B). To confirm that the cross-coupling reaction is initiated by photo-induced Ni−C bond homolysis, mixtures of **1** (1 equiv) and TEMPO radical trap (2 equiv) in THF were monitored by GCMS under conditions with and without blue-light irradiation.[42] GCMS analysis showed the formation of the TEMPO-Me adduct only when the reaction mixture was irradiated with light (Figure 3C). A possible accelerating effect of light beyond the catalyst initiation process was probed with light-on and light-off studies (Figure 3D). A mixture of **1** (5 mol %), **4**, and Me_2Zn was irradiated with blue light for 5 or 10 min, after which the light was turned off, and the reaction was monitored by LCMS in the dark at room temperature for 5 h (Figure 3D). Although the reaction in which the light was turned off after 5 min (curve II) appeared to initially progress more slowly than the one in which the light was left on for the duration (curve III), both reactions proceeded to full conversion of **7** within 5 h (Figure 3D). The rate of the reaction when the light was turned off after 10 min (curve IV) was similar to the one where the light irradiation was maintained for 5 h (curve III). This result suggests that once the dimethylNi^{II} precatalyst is fully activated, the reaction is no longer dependent on the presence of light. In the complete absence of light, the reaction did not proceed (curve I). The feasibility of photo-induced Ni−C bond homolysis is also supported computationally with DFT calculations of the bond dissociation free energies (BDFE). DFT calculations revealed that $[(dtbbpy)Ni(Me)_2]$ (**1**) possesses a low triplet−singlet (T_1-S_0) free energy gap of +18.6 kcal/mol (Figure 3E).[43] Estimation of the BDFE of triplet excited state **1*** yielded ΔG_{diss} = +17.6 kcal/mol[43,49] for the homolysis of the first Ni−C bond and a negligible value of ΔG_{diss} = +0.1 kcal/mol for the homolysis of the second Ni−C bond, which could spontaneously homolyze to give a solvated Ni(0) species.[50]

It is noteworthy that the calculated BDFE for the Ni−C(sp^3) bond is lower than the values reported for Ni−Ar, Ni−Cl, and Ni−Br bonds.[43] The facile scission of both Ni−C(sp^3) bonds supports the mechanistic proposal for the light-induced activation of **1** (Figures 1F and 2). Lastly, critical control experiments with a Ni^{0}-precursor, $Ni(COD)_2$, in the absence of light, were conducted. As illustrated in Figure 3F, the combination of $Ni(COD)_2$/dtbbpy mediated the cross-coupling of **4** with Me_2Zn and Et_2Zn in the dark to deliver **7** and **41**, respectively, in quantitative yield. These control experiments, together with the one in Figure 3B, provide experimental support for the Ni^{0}−Ni^{II}−Ni^{II}−Ni^{0} catalytic cycle illustrated in Figure 2B. Collectively, these experimental and computational studies support a mechanism in which light induces the Ni−C bond homolysis of a dimethylNi^{II} complex to produce an active Ni^{0}-species that mediates the cross-coupling reaction in the absence of light.

In summary, we have unveiled the hitherto elusive reaction potential of stabilized dimethylNi^{II} complexes that can be activated by irradiation with visible light. We studied the feasibility of di-*tert*-butylbipyridine and bipyridine-stabilized Ni^{II}Me$_2$ complexes **1** and **2** in promoting C−C cross-coupling reactions via a light-initiated, Ni-catalyzed protocol that

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excludes the use of a photoredox catalyst. Complexes **1** and **2** were unmasked by light-induced Ni−C bond homolysis to produce active Ni^0 species and reactive methyl radicals, which led to the development of a practically simple and broadly useful methylative cross-coupling protocol. We also demonstrated that these stabilized dimethylNi^{II} complexes can be used either stoichiometrically or catalytically to promote C_{sp}^2−C_{sp}^3 cross-coupling reactions, allowing access to sp^3-rich molecular scaffolds. When a dimethylNi^{II} complex is employed for cross-coupling, light may initiate a dark stoichiometric reaction or promote a catalytic cycle. Furthermore, we have demonstrated that these new protocols are amenable to the late-stage functionalization of drug-like and advanced-stage molecules from their aryl bromide precursors. Experimental mechanistic studies support a Ni^0−Ni^{II}−Ni^{II}−Ni^0 catalytic cycle for the C_{sp}^2−C_{sp}^3 cross-coupling, while Ni^0−Ni^{II}−Ni^{III}−Ni^I is proposed for the stoichiometric methylative cross-coupling. While it is believed that Ni^0-species mediates the cross-coupling when a dimethylNi^{II} precatalyst is employed, the involvement of Ni^I−Me species in the cross-coupling process cannot be ruled out. The visible-light activation of stabilized dialkylNi^{II} offers a novel and practically convenient mechanism for initiating and promoting Ni-catalyzed processes, especially where a Ni^{II}-precatalyst is employed. We anticipate that the potential of stabilized dimethylNi^{II} complexes unveiled in this report will further the exploration of other dialkylNi^{II} complexes, and the new mechanistic insights provided will add to the understanding of Ni-catalyzed C_{sp}^2−C_{sp}^3 cross-coupling reactions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.2c04213.

Experimental procedures; characterization data; ^1H, ^{13}C, and ^{19}F NMR spectra; and computational methods (PDF)

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(49) Full geometry optimizations were carried out with Gaussian 09 at the (U)M06L/def2-SVP level of theory. See Supporting Information for references.

(50) Microsolvation by a varying number of THF molecules was explored for all Ni complexes and considered when the solvent association was thermodynamically favored ($\Delta G < 0$).

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